BIOT 6900 · AI-Powered Biotech Workflows · Computational Lab 6

Polygenic Risk Score Portability on Real 1000 Genomes Data

Mohammed Sharukh · Fall 2026 · Repository: github.com/MohammedSharukh7/biot6900 (module6/)

1. Overview

I applied the published late-onset Alzheimer's disease score **PGS000334 (GRSfull_22)** from the PGS Catalog to real 1000 Genomes phase 3 genotypes for 2,336 people in five super-populations. The workflow harmonized alleles, computed a standardized PRS, measured how the PRS distribution and a single top-10% cutoff differ by ancestry, evaluated discrimination (AUC) and calibration (EUR-fitted model) within each group, and tested incremental value over age. Section 6 applies the same workflow to a pancreatic ductal adenocarcinoma (PDAC) score, my Part I disease.

The phenotype is simulated. 1000 Genomes has no disease labels. The PRS and its distribution across ancestries are computed from real genotypes and published weights. AUC, calibration and the age comparison use a simulated case label, so they are illustrative only (simulated case fraction ≈ 30%, far above real Alzheimer's prevalence).

2. Proof of real data (Section 2 output)

```
source      : REAL
people      : 2336
score size  : 22 variants
by ancestry : {'AFR': 500, 'EAS': 500, 'EUR': 500, 'SAS': 489, 'AMR': 347}
  genotypes.csv: sha256 9cf41a1f
  variants.csv: sha256 f7a87d3f
  score.csv: sha256 7f80e8ff
  meta.csv: sha256 6cc55b6e
```

3. Methods

Genotype dosages count the ALT allele. For each score variant, the effect allele (uppercased) was compared with REF and ALT: **keep** if it equals ALT, **flip** the dosage to 2 − dosage if it equals REF, and **drop** if it equals neither. Score variants absent from the panel were counted separately. The PRS is the matrix product of aligned dosages and weights, standardized across the whole cohort. A single cutoff was set at the pooled 90th percentile. AUC was computed overall and within each ancestry. For calibration, a logistic model of phenotype on PRS was fitted in EUR only and applied to everyone, using the provided five-bin quantile helper. Incremental value used 5-fold cross-validated AUC for age alone versus age + PRS.

4. Results

Harmonization (3.1). All 22 score variants were found in the panel (0 not in panel). 22 were used, 13 needed flipping because the published effect allele is the REF allele, and 0 were dropped for allele mismatch. So there is no could-not-score gap: every weight in the score is applied.

Score variants	Found	Not in panel	Used	Flipped	Dropped
22 (PGS000334)	22	0	22	13	0

Table 1. Allele harmonization.

Ancestry	n	Mean PRS (SD)	% above top-10% cutoff	AUC (cases)	Mean predicted risk	Observed risk	Gap (pred − obs)
EUR	500	0.04	7.8	0.672 (157)	0.314	0.314	+0.000
AFR	500	0.74	27.8	0.493 (187)	0.418	0.374	+0.044
EAS	500	−0.47	2.2	0.575 (140)	0.246	0.280	−0.034
SAS	489	−0.25	4.3	0.611 (125)	0.273	0.256	+0.017
AMR	347	−0.10	6.3	0.555 (92)	0.293	0.265	+0.028
All	2336	0.00	10.0	0.599 (701)	—	—	—

Table 2. Distribution shift (3.3), discrimination (3.4) and calibration-in-the-large of the EUR-fitted model (3.5). Cutoff = 1.493 SD (pooled 90th percentile).

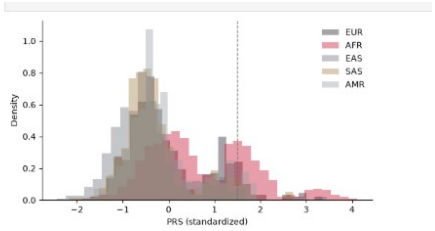


Figure 1. PRS by ancestry; dashed = top-10% cutoff.

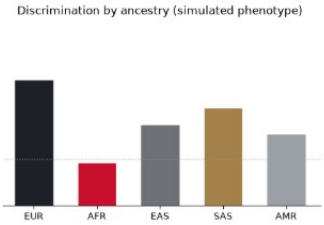


Figure 2. AUC by ancestry.

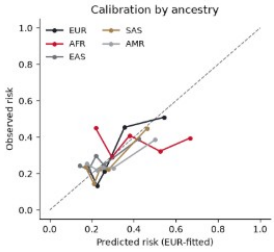


Figure 3. Calibration, EUR-fitted model.

Incremental value (3.6). Cross-validated AUC was 0.562 for age alone and 0.618 for age + PRS, an increase of **+0.056**.

Allele frequencies (extra, 3.3b). Effect-allele frequencies differ widely across groups (Table 3). The largest spread is 0.50 (rs9381563).

Variant (effect allele)	Weight	EUR	AFR	EAS	SAS	AMR
rs9381563	−0.08	0.678	0.308	0.806	0.591	0.644
rs593742	+0.07	0.680	0.695	0.212	0.459	0.548
rs11136000 (CLU)	−0.11	0.386	0.581	0.219	0.282	0.357
rs1476679 (ZCWPW1)	+0.09	0.700	0.984	0.650	0.766	0.774
rs6656401 (CR1)	+0.14	0.174	0.008	0.036	0.061	0.079
rs429358 (APOE; T = non-ε4)	−1.13	0.846	0.738	0.914	0.913	0.896

Table 3. Effect-allele frequency by ancestry for selected score variants (from the 3.3b output).

5. Interpretation

Q1. Why does the European-derived score lose accuracy in other ancestries?

Most score variants are GWAS *tag* SNPs, and their weights were estimated in European samples. A weight captures the effect of a causal variant *as seen through its LD with the tag in Europeans*. Two non-biological factors break this outside EUR.

- **Linkage disequilibrium differs.** African-ancestry populations are older and more diverse, so LD blocks are shorter and a European tag is a weaker proxy for the causal variant. East Asian, South Asian and admixed American groups each have their own LD patterns. When tag–causal correlation drops, the tag carries the European weight but less of the risk signal.
- **Allele frequencies differ.** Table 3 shows effect-allele frequencies that change by up to 0.50 between groups. A frequency shift moves the mean PRS (AFR +0.74 SD, EAS −0.47 SD; Table 2) without any change in risk, and it changes which variants drive the ranking. A variant almost absent in a group, such as the CR1 allele (0.008 in AFR vs 0.174 in EUR), contributes nothing to ranking there. Risk variants that are common only outside Europe were never discovered, so they are missing from the score.

The result tracks this: AUC is highest in the training ancestry (EUR 0.672), lower in SAS (0.611), EAS (0.575) and AMR (0.555), and at chance in AFR (0.493). The disease is the same; it is the score's statistical proxies that fail to transfer.

Q2. One threshold, many groups: is a single top-10% cutoff fair?

No. The pooled cutoff (1.493 SD) flags **27.8%** of AFR participants but only **2.2%** of EAS, a 12.6-fold difference, with EUR 7.8%, AMR 6.3% and SAS 4.3%. About 139 of the ≈234 flagged people are African-ancestry, roughly 59% of the high-risk group, although AFR is 21% of the cohort. The shift appears in the real PRS before any phenotype is used, so it reflects allele frequencies, not demonstrated risk. In AFR the score also does not rank risk (AUC 0.493), so flagged AFR people are no more likely to be cases than unflagged ones. Meanwhile EAS observed risk (0.280) exceeds the model's prediction (0.246), and few EAS people are ever flagged.

A clinic applying this cutoff unchanged would over-refer African-ancestry patients for follow-up, counselling and testing, with anxiety and cost but no gain in detection. It would under-refer East Asian patients whose risk is understated. The cutoff also depends on the clinic's ancestry mix: changing the mix changes who is flagged. Safer options are ancestry-aware standardization, for example z-scoring the PRS against a matched reference distribution or genetic principal components, recalibrating absolute risk within each group, using multi-ancestry scores, or not reporting the score in groups where it has not been validated.

Ethical implications. A tool that works best for people of European ancestry and misclassifies others risks widening existing health disparities, because most GWAS participants are European. APOE-containing results are especially sensitive: they concern an incurable disease and are shared with relatives. In the US, GINA covers health insurance and employment but not life, disability or long-term-care insurance. Any use needs informed consent, genetic counselling, a right not to know, and transparent reporting of which populations the score was validated in.

Q3. Discrimination vs calibration: what did each show, and where did they disagree?

Discrimination (3.4) asks whether the PRS *ranks* cases above controls. It is best in EUR and fails in AFR. Calibration (3.5) asks whether predicted probabilities match observed risk. The EUR-fitted model is exact in EUR by construction (0.314 vs 0.314) and sits near the diagonal. Elsewhere it drifts: it over-predicts in AFR (+0.044), AMR (+0.028) and SAS (+0.017) and under-predicts in EAS (−0.034).

- **AFR:** the calibration-in-the-large gap is modest, so the *average* prediction looks nearly honest, but AUC is at chance and the calibration curve is flat and erratic. The highest-predicted AFR bins have observed risk well below their predictions. A good mean can hide a useless ranking; only the curve shape and the AUC expose it.
- **SAS:** this group has the best non-EUR ranking (AUC 0.611) yet small miscalibration. The ranking transfers better than the absolute numbers.
- **EAS:** discrimination is moderate (0.575), but risk is systematically understated, so a patient would be told a lower risk than reality.

The two measures react to different things. AUC is unchanged by a shift in mean PRS, while calibration is not: the AFR shift of +0.74 SD feeds directly into inflated predicted risk. Both are needed before a score is trusted in a group.

Q4. Clinical utility and limits

Where it could help. In European-ancestry adults the score adds modestly to age (cross-validated AUC 0.562 → 0.618). Much of that comes from APOE, the dominant weight. It could support research uses such as enriching prevention-trial cohorts, or act as one input alongside age and family history. APOE ε4 status is already considered before anti-amyloid antibody therapy because carriers face higher risk of amyloid-related imaging abnormalities.

Where it should not be used. It should not be used to diagnose an individual, as a population screening threshold, or for any decision in African-ancestry patients with this score, given AUC at chance. It should not be used for insurance or employment decisions, or disclosed predictively without counselling.

What is needed before deployment. The phenotype here is simulated, and its 30% case fraction is unrealistic, so these AUC and calibration values cannot be used clinically. Deployment would need validation against real,

clinically or neuropathologically confirmed Alzheimer's diagnoses (ideally incident cases) in every ancestry where it will be used, recalibration to local age-specific incidence, an absolute-risk model with age, sex and family history, reclassification metrics, and prospective evaluation. Suitable real data include the Alzheimer's Disease Sequencing Project and Alzheimer's Disease Genetics Consortium case–control data (which include African American cohorts), NACC clinical and neuropathology records, and diverse biobanks with linked diagnoses such as All of Us, the Million Veteran Program, Biobank Japan and H3Africa. UK Biobank is useful but mostly European.

Q5. Artifact versus biology: the APOE ε4 result in AFR

The near-chance AFR AUC is a property of how this dataset's phenotype was simulated. According to the data README, the simulated label depends on an LD proxy for ε4 that is almost absent in African-ancestry samples, so ε4 barely contributes to simulated case status in AFR. The real genotypes show the opposite of "unimportant": rs429358 is the ε4-defining SNP itself, and the ε4 allele (1 − 0.738) is **more** common in AFR (0.26) than in EUR (0.15), EAS (0.09), SAS (0.09) or AMR (0.10). Real association studies report that ε4 remains a risk allele in African-ancestry populations, though often with a smaller per-allele effect that has been linked to the ancestral haplotype surrounding APOE. A low AUC in one group can be produced by the data-generating design or by LD, without any absence of biological effect. Simulated labels cannot support a biological conclusion.

6. Extension: the same workflow on my Part I disease (PDAC)

The Canvas panel contains only Alzheimer's loci, so no pancreatic-cancer variant could be scored on it. I therefore built a PDAC dataset for the same 2,336 people. I used score **PGS000725** (PRS_Pancreas; Jia et al., JNCI Cancer Spectrum 2020; 22 genome-wide-significant variants, GWAS and evaluation 100% European, reported AUROC 0.639 for incident pancreatic cancer). Genotypes were retrieved from 1000 Genomes phase 3 via the Ensembl REST API (script *build_pdac_data.py*). ALT was set to the non-REF allele actually observed in 1000 Genomes, because dbSNP lists extra alleles at sites such as rs3790844 (NR5A2) and rs9543325 (13q22.1).

Quality control. 21/22 variants were retrieved with no missing samples. rs16986825 (22q12.1) is reported as not in panel after repeated server errors. EUR effect-allele frequencies matched the PGS-reported frequencies for all 21 variants within ±0.035, confirming strand and orientation, including the palindromic A/T SNPs rs73328514 (0.896 vs 0.90) and rs10094872 (0.350 vs 0.36).

Ancestry	Mean PRS (SD)	% above top-10% cutoff	Largest effect-allele-frequency differences
EUR	[ ]	[ ]	[from PDAC 3.3b output]
AFR	[ ]	[ ]	
EAS	[ ]	[ ]	
SAS	[ ]	[ ]	
AMR	[ ]	[ ]	

Table 4. PDAC PRS (PGS000725) distribution shift and single-cutoff fractions in the same cohort.

[PDAC interpretation: 2–3 sentences comparing the PDAC shift with the Alzheimer's shift — to be completed from the extension notebook output.]

AUC and calibration were deliberately not computed for PDAC. There are no PDAC labels for 1000 Genomes, and a label simulated from this same score would build the answer in. Clinically, PDAC is rare and its PRS discrimination is modest, so a PRS alone is unsuitable for population screening. Surveillance programmes instead target people with germline predisposition or strong family history, and any PRS role would need validation in multi-ancestry cohorts with confirmed PDAC diagnoses.

7. Limitations

Simulated phenotype (AUC, calibration and incremental value are illustrative); a small panel (22 variants); modest group sizes, with AMR n = 347 and five-bin calibration curves that are noisy; super-population labels that compress substantial within-group diversity, especially in admixed AMR; and no adjustment for genetic principal components.

References

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2. PGS Catalog. PGS000725 (PRS_Pancreas). Jia G et al. JNCI Cancer Spectrum (2020). https://www.pgscatalog.org/score/PGS000725
3. 1000 Genomes Project Consortium. A global reference for human genetic variation. Nature 526, 68–74 (2015).
4. Martin AR et al. Clinical use of current polygenic risk scores may exacerbate health disparities. Nature Genetics 51, 584–591 (2019).
5. Ensembl REST API, variation endpoint (1000 Genomes phase 3 genotypes). https://rest.ensembl.org
6. BIOT 6900 Module 6 Lab Guide Part 3 and graded starter notebook; data README (Canvas).